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Site NDE Procedure - Ultrasonic Thickness Measurement, ATU-100

1.0 Scope and Purpose

This procedure establishes the requirements for ultrasonic thickness measurement and supplemental NDE within the ATU-100. It incorporates critical updates to address external corrosion mechanisms identified during recent inspections. This document supersedes QSCP-NDT-001 Rev 1 and integrates findings from the root cause analysis regarding localized wall loss at fittings.

The primary objective is to ensure accurate data collection for the calculation of remaining life and corrosion rates. Per site rules, the retirement thickness is defined as the line t-min. Any reading recorded below t-min must be formally documented as an NCR, following the precedent set by QSCP-NCR-2010-007 where wall loss was identified on 4-RA-104.

2.0 Responsibilities

Douglas Ferreira is responsible for the technical oversight of NDE activities, ensuring that all technicians utilize gauges with waveform displays and perform scanning at elbows and tees rather than spot readings. The Inspection Supervisor must verify that CMLs are correctly located at fittings to capture maximum degradation as required by the revised site standards.

Robert Hendricks is responsible for reviewing inspection data to determine the next inspection interval based on calculated remaining life. The Unit Engineer ATU-100 ensures that the inspection program adheres to the class maxima, specifically maintaining the 3-year thickness interval for the regenerator overhead circuit and the 5-year or 10-year intervals for other amine and utility services.

3.0 Corrosion Under Insulation (CUI) Program

In accordance with QSCP-IP-ATU100 Rev 2, all insulated lines operating within the CUI range are subject to specific NDE requirements. This program was initiated to prevent failures similar to those documented in QSCP-NCR-2010-007, where insulation stripping revealed external CUI below t-min. The program mandates that these circuits receive profile RT or the installation of insulation windows every 5 years.

Technicians must coordinate insulation removal and reinstatement to facilitate direct UT scanning. Where insulation removal is not feasible, profile radiography shall be employed to screen for external wall loss. All inspection-port requirements must be met to ensure repeatable measurements at established CMLs during subsequent inspection cycles.

4.0 Data Analysis and Inspection Intervals

Corrosion rates are determined by selecting the larger of the long-term rate or the short-term rate, provided the latest readings are at least 3 years apart. The remaining life is calculated by dividing the difference between the current thickness and t-min by the determined rate. The next thickness reading must be scheduled at no more than half the remaining life, not to exceed the class maximum.

The QSCP-IP-ATU100 Rev 2 defines the class maxima for the unit. Class 1 services such as sour gas require thickness and external visual inspections every 5 years. Class 2 services, including amine and treated gas, are set at 10 years for thickness and 5 years for visual. Class 3 utility lines are set at 10 years for both thickness and visual inspections.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2011-01-17	_____
Approved	Robert Hendricks	Unit Engineer ATU-100, P.Eng.	2011-01-17	_____